

Melody Slot Machine HD

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Figure 1: Screenshots of Melody Slot Machine HD

ABSTRACT

Melody Slot Machine HD is an application that allows users to experience generating melodies using musical structures while playing slot machines. To make it possible, we use the Generative Theory of Tonal Music (GTTM) to analyze the melody structure and generate variations of the melody. All variation melodies created on the basis of GTTM have a similar musical structure, so even if the melody switches to another variation in the middle, the overall structure will not collapse. It is possible to see the player on the display playing a new combination of a melody generated interactively by the user's manipulation.

CCS CONCEPTS

• Applied computing → Sound and music computing.

KEYWORDS

GTTM, time-span tree, melody morphing, music structure

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1 INTRODUCTION

Melody Slot Machine HD is an iPad application that allows you to replace segments of a melody with other variations and watch the performer on the display play the resulting melody. When you

shake your iPad or press the shuffle button, the symbols on the reels will rotate and stop randomly.

We used a melody morphing method based on the Generative Theory of Tonal Music (GTTM) to generate melodies for variations, enabling natural melodies to be generated even when one melody segment is randomly replaced by another variation [Hamanaka et al. 2009, 2023; Lerdahl and Jackendoff 1983].

2 MELODY SLOT MACHINE HD

Figure 1a is a screenshot of Melody Slot Machine HD in which the symbols represent changes in melody variations. For example, the musical note symbol means that the same variation will continue. The cherry symbol indicates that the variations will change one after another.

Pressing the mode switch buttons on the left and right of the screen displays the grid tile screen, and you can check and change the variations in the entire song (Fig. 1b). If you use the grid to change the combination of variations, the symbols on the slot screen will also change accordingly.

Press the mode switch buttons or hold the iPad vertically to display the performer screen (Fig. 1c, d). This shows a performer playing new combinations of melodies determined by the slots or grids. Short video clips of the performer generated by AI are sandwiched into recorded videos of an actual performer, so the performer moves seamlessly.

Melody Slot Machine HD has two features: **Composition experience**: Melody Slot Machine HD can generate a huge number of combinations of melodies. For example, in Fig. 1b, 6 variation melodies are divided into 16 melody segments in the time direction, so there are $6^{16} = 2,821,109,907,456$ combinations of melodies. Therefore, the user can feel as if he or she is composing a melody, because a combination of melodies that the user has not yet heard is generated by his/her own operation.

Performer control experience: The performer display inserts an interpolated video generated by AI at the joint of the live-action video of the performer in each segment. For example, in Fig. 1b,

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we created $6^2 \times 16 = 576$ interpolated videos. Since the performance of the live-action changes in accordance with the user's operation, the user can feel as if he/she is controlling the performer himself/herself.

3 MELODY MORPHING METHOD

The variation melodies are generated by a melody morphing method based on GTTM. The time-span tree in Fig. 2 is the result of analyzing Melody A on the basis of GTTM. Reduced melodies can be extracted by cutting this time-span tree with a horizontal line and omitting the notes connected below the line. Melody reduction with GTTM is the absorption of structurally unimportant notes by structurally important notes.

In melody morphing, we use the primitive operations of the subsumption relation (written as $\sqsubseteq$), meet (written as $\sqcap$), and join (written as $\sqcup$). Subsumption represents the relation by which “an instantiated object” $\sqsubseteq$ “an abstract object.” The meet operator extracts the largest common part or the most common information of the time-span trees of two melodies in a top-down manner. The join operator joins two time-span trees in the top-down manner as long as their structures are consistent.

Figure 3 is an overview of the melody morphing method. By using the time-span trees σ_A and σ_B from Melodies A and B, we can calculate the most common information, $\sigma_A \sqcap \sigma_B$, which are the essential parts of melody A, as well as those of Melody B. The meet operations $\sigma_A \sqcap \sigma_B$ are abstracted from σ_A and σ_B , and those discarded notes are regarded as the difference information of σ_A and σ_B (Fig. 3a). We consider that there are features without the other melody in the difference information of σ_A and σ_B . Therefore, we need a method for smoothly increasing or decreasing these features. The melody divisional reduction method can abstract the notes of the melody in the differential branch of the time-span tree (Fig. 3b). We can acquire the morphed melody between σ_A and σ_B by using the join operator to combine Melodies σ_{α_i} and σ_{β_i} , which are the results of the divisional reduction (Fig. 3c).

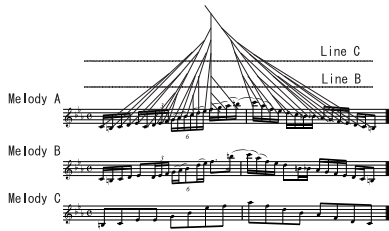


Figure 2: Time-span tree

4 RELATED WORK

The purpose of the *musical dice game* is to combine the melody selected by the dice in order [Hedges 1978]. Melody Slot Machine HD was inspired by the dice game idea in terms of changing the melody. Melodies in the dice game are designed very elaborately so that they sound natural in any order. Our melody morphing method uses the result of music structural analysis from the GTTM. Even if the melody is changed, the essential structure is the same and only ornamentation notes change.

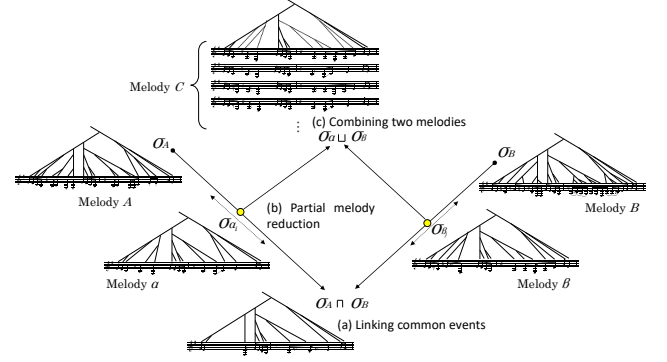


Figure 3: Overview of melody morphing method

At the start of app development, because of the low accuracy of the time-span tree analyzer of the GTTM, composers manually performed time-span tree analysis to generate morphing melodies [Hamanaka et al. 2006; Hirata et al. 2022; Kanamori et al. 2014]. The introduction of deep learning has dramatically improved the analysis performance, making it possible to automate from time-span tree analysis to morphing melody generation [Hamanaka et al. 2021, 2017a,b].

5 CONCLUSION

This paper describes an iPad application called Melody Slot Machine HD, which uses a melody morphing method based on GTTM. The application download and app introduction video can be accessed at the following URL:

<https://gttm.jp/hamanaka/en/melodyslotmachinehd/>.

Currently, two songs have been released, and we plan to add more content in the future.

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